

4.6.4

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October 3, 2025

Question

Find the equation of a line passing through the point $(2, 3, 2)$ and parallel to the line $\mathbf{r} = (-2\hat{i} + 3\hat{j}) + \lambda(2\hat{i} - 3\hat{j} + 6\hat{k})$. Also, find the distance between these two lines.

Step 1: Equation of the Required Line

The vector equation of a line is $\mathbf{r} = \mathbf{a}_2 + \mu\mathbf{b}$.

The line passes through the point $(2, 3, 2)$, so the position vector is:

$$\mathbf{a}_2 = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} \quad (1)$$

The line is parallel to the given line, so they share the same direction vector:

$$\mathbf{b} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \quad (2)$$

The equation of the required line is:

$$\mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \quad (3)$$

Step 2: Distance Calculation

The distance is found by identifying points $\mathbf{P} = \mathbf{a} + p_1\mathbf{d}$ and $\mathbf{Q} = \mathbf{b} + p_2\mathbf{d}$ on each line such that the vector $\mathbf{Q} - \mathbf{P}$ is orthogonal to the direction $\mathbf{d}$. The problem defines the two parallel lines. We can represent them with the following vectors:

$$\mathbf{a} = \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix} \quad (\text{Point on Line 1}) \quad (4)$$

$$\mathbf{b} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} \quad (\text{Point on Line 2}) \quad (5)$$

$$\mathbf{d} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \quad (\text{Common Direction}) \quad (6)$$

Step 2: Distance Calculation

The orthogonality condition is $\mathbf{d}^\top (\mathbf{Q} - \mathbf{P}) = 0$. Expanding this gives the equation for the parameters:

$$(\mathbf{d}^\top \mathbf{d})(p_2 - p_1) + \mathbf{d}^\top (\mathbf{b} - \mathbf{a}) = 0 \quad (7)$$

We calculate the scalar products:

$$\mathbf{d}^\top \mathbf{d} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}^\top \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} = 4 + 9 + 36 = 49 \quad (8)$$

$$\begin{aligned} \mathbf{d}^\top (\mathbf{b} - \mathbf{a}) &= \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}^\top \left(\begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} - \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix} \right) \\ &= \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}^\top \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} = 8 + 0 + 12 = 20 \end{aligned} \quad (9)$$

Step 2: Distance Calculation

Substituting these values back:

$$49(p_2 - p_1) + 20 = 0 \quad (10)$$

$$p_2 - p_1 = -\frac{20}{49} \quad (11)$$

$$\implies p_1 - p_2 = \frac{20}{49} \quad (12)$$

We need one pair of values that satisfy this. We can simply take $\mathbf{p}_2 = \mathbf{0}$, which gives $\mathbf{p}_1 = \mathbf{20/49}$.

Using these parameter values, we find the specific points $\mathbf{P}$ and $\mathbf{Q}$.

$$\mathbf{P} = \mathbf{a} + p_1 \mathbf{d} = \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix} + \frac{20}{49} \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} = \begin{pmatrix} -58/49 \\ 87/49 \\ 120/49 \end{pmatrix} \quad (13)$$

$$\mathbf{Q} = \mathbf{b} + p_2 \mathbf{d} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} \quad (14)$$

Step 2: Distance Calculation

The vector connecting these points is:

$$\mathbf{P} - \mathbf{Q} = \begin{pmatrix} -58/49 - 98/49 \\ 87/49 - 147/49 \\ 120/49 - 98/49 \end{pmatrix} = \frac{1}{49} \begin{pmatrix} -156 \\ -60 \\ 22 \end{pmatrix} \quad (15)$$

The distance is the norm of this vector, $\|\mathbf{P} - \mathbf{Q}\|$.

$$\begin{aligned} d^2 &= \left\| \frac{1}{49} \begin{pmatrix} -156 \\ -60 \\ 22 \end{pmatrix} \right\|^2 \\ &= \left(\frac{1}{49} \right)^2 \left((-156)^2 + (-60)^2 + (22)^2 \right) \\ &= \frac{1}{2401} (24336 + 3600 + 484) = \frac{28420}{2401} = \frac{580}{49} \end{aligned} \quad (16)$$

$$d = \sqrt{\frac{580}{49}} = \frac{\sqrt{4 \times 145}}{7} \quad (17)$$

```
#include <stdio.h>
#include <math.h>

int main() {
    double a1[3] = {-2, 3, 0};
    double a2[3] = {2, 3, 2};
    double b[3] = {2, -3, 6};
    double da[3], v[3];

    for (int i = 0; i < 3; i++) da[i] = a2[i] - a1[i];
```



```
v[0] = b[1]*da[2] - b[2]*da[1];  
v[1] = b[2]*da[0] - b[0]*da[2];  
v[2] = b[0]*da[1] - b[1]*da[0];  
  
double vnorm = sqrt(v[0]*v[0] + v[1]*v[1] + v[2]*v[2]);  
double bnorm = sqrt(b[0]*b[0] + b[1]*b[1] + b[2]*b[2]);  
  
double d = vnorm / bnorm;  
printf("Distance = %lf\n", d);  
  
return 0;  
}
```

Python + C Code

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Load compiled shared library
lib = ctypes.CDLL('./line_distance.so')
lib.main()

# Given data
a1 = np.array([-2, 3, 0])
a2 = np.array([2, 3, 2])
b = np.array([2, -3, 6])

# Compute distance
da = a2 - a1
v = np.cross(b, da)
d = np.linalg.norm(v) / np.linalg.norm(b)
print("Distance =", d)
```

```
# Projection of da onto b to find perpendicular foot
```

```
proj = (np.dot(da, b) / np.dot(b, b)) * b
```

```
perp_vec = da - proj
```

```
p1 = a1 + proj
```

```
p2 = a2
```

```
# Plot
```

```
fig = plt.figure()
```

```
ax = fig.add_subplot(111, projection='3d')
```

```
# Line 1 and Line 2
```

```
t = np.linspace(-2, 2, 10)
```

```
line1 = a1 + np.outer(t, b)
```

```
line2 = a2 + np.outer(t, b)
```

```
ax.plot(line1[:,0], line1[:,1], line1[:,2], 'r', label='Line 1')
```

```
ax.plot(line2[:,0], line2[:,1], line2[:,2], 'b', label='Line 2')
```

```
# Points with vector labels
ax.scatter(*a1, color='red')
ax.text(a1[0], a1[1], a1[2], f"A1 {a1}", color='red', fontsize
        =10)
ax.scatter(*a2, color='blue')
ax.text(a2[0], a2[1], a2[2], f"A2 {a2}", color='blue', fontsize
        =10)

# Perpendicular distance line
ax.plot([p1[0], p2[0]], [p1[1], p2[1]], [p1[2], p2[2]], 'g--',
        label=f'Distance = {round(d,3)}')

ax.legend()
plt.show()
```

Python Plotting Code

```
import numpy as np
import matplotlib.pyplot as plt

# Data
a1 = np.array([-2, 3, 0])
a2 = np.array([2, 3, 2])
b = np.array([2, -3, 6])

# Distance calculation
da = a2 - a1
v = np.cross(b, da)
d = np.linalg.norm(v) / np.linalg.norm(b)
print("Distance =", d)

# Projection for perpendicular line
proj = (np.dot(da, b) / np.dot(b, b)) * b
p1 = a1 + proj
p2 = a2
```

Python Plotting Code

```
# Plot
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

# Lines
t = np.linspace(-2, 2, 10)
line1 = a1 + np.outer(t, b)
line2 = a2 + np.outer(t, b)

ax.plot(line1[:,0], line1[:,1], line1[:,2], 'r', label='Line 1')
ax.plot(line2[:,0], line2[:,1], line2[:,2], 'b', label='Line 2')

# Points with labels
ax.scatter(*a1, color='red')
ax.text(a1[0], a1[1], a1[2], f"A1 {a1}", color='red', fontsize
        =10)
ax.scatter(*a2, color='blue')
ax.text(a2[0], a2[1], a2[2], f"A2 {a2}", color='blue', fontsize
        =10)
```

Python Plotting Code

```
# Perpendicular distance line
ax.plot([p1[0], p2[0]], [p1[1], p2[1]], [p1[2], p2[2]], 'g--',
        label=f'Distance = {round(d,3)}')

# Direction vectors (as arrows from A1 and A2)
ax.quiver(a1[0], a1[1], a1[2], b[0], b[1], b[2], color='red',
          arrow_length_ratio=0.1, linewidth=1.5)
ax.text(a1[0]+b[0]*0.5, a1[1]+b[1]*0.5, a1[2]+b[2]*0.5, "b",
        color='red')

ax.quiver(a2[0], a2[1], a2[2], b[0], b[1], b[2], color='blue',
          arrow_length_ratio=0.1, linewidth=1.5)
ax.text(a2[0]+b[0]*0.5, a2[1]+b[1]*0.5, a2[2]+b[2]*0.5, "b",
        color='blue')

# Make plot readable
ax.legend()
ax.set_xlabel("X")
ax.set_ylabel("Y")
```

Plot of the Parallel Lines

